MASTER SOFTWARE REQUIREMENTS, QUANTITATIVE ARCHITECTURE & SDLC SPECIFICATION

Bharat Fintech & AI Market Intelligence Platform

End-to-End Corporate Engineering Blueprint for an Institutional-Grade, Multi-Page Indian Capital Markets Encyclopedia, Quantitative Derivatives & Fundamental Calculation Engine, and Multi-Agent AI Research System

Target Engineering Profile: Python Backend Engineer | AI Engineer | GenAI Engineer | Quantitative Full-Stack Developer • **Market Scope:** NSE, BSE, NSE IX (GIFT Nifty), MCX, AMFI, RBI, SEBI • **Infrastructure Budget:** Zero-Cost (₹0) Free-Tier Production Stack

1. Executive Overview & Core Architectural Philosophy

The Bharat Fintech & AI Market Intelligence Platform is a production-ready, multi-page web application engineered to serve as an all-in-one institutional encyclopedia, fundamental screener, derivatives (F&O) analytics terminal, and quantitative calculation engine for the Indian capital markets. When a user searches for any listed Indian company, NSE/BSE Index, GIFT Nifty or Indian ADR pre-market ticker, Stock or Index Futures & Options chain, upcoming or historical IPO, Mutual Fund, Exchange-Traded Fund (ETF), Government or Corporate Bond, Commodity (Gold and Silver), or REIT/InvIT, the platform delivers five-year and twenty-quarter historical depth, custom backend financial computations, interactive investment calculators, corporate event tracking, and factual AI-driven market intelligence.

Core Engineering Guardrails & Non-Negotiable Product Rules:

- **Contextual Missing-Data Architecture (Zero Raw Nulls):** The backend never exposes a bare `null` or unexplained dash to the frontend. Every financial metric pairs a numerical value with a standardized status code (`REPORTED`, `LISTED-RECENTLY-UNAVAILABLE`, `NOT-APPLICABLE-FOR-SECTOR`, or `NOT-DISCLOSED-BY-COMPANY`) and a human-readable explanation (for example, *"Listed on 10 Nov 2024 — 5-Year historical data does not exist yet"* or *"Inventory Turnover is Not Applicable for Banking/NBFC Sector"*).
- **On-Demand Lazy-Loaded Peer Benchmarking with Persistent Session State:** Peer comparison tables never load automatically on initial page render. Every asset page places an explicit **+ Compare with Peers** action button in the top sticky header and beside the interactive calculators. Clicking the button lazily triggers the peer comparison API and opens the interactive comparison workspace, reducing initial page payload and database load by over 60 percent. Once opened by the user, the active peer comparison state persists in session storage so switching across sub-tabs (Ratios, Cash Flow, Shareholding, or Portfolio Overlap) compares the selected peers seamlessly.
- **True "Net-in-Pocket" Statutory Charge & Tax Transparency:** Every equity, F&O, mutual fund, ETF, and bond calculator includes an automated statutory deduction engine calculating exact Securities Transaction Tax (STT), NSE/BSE Exchange Turnover Charges, SEBI Turnover Fees, Stamp Duty, Depository Participant (DP) charges, and 18% GST so users see exact breakeven points and net take-home returns.

- **Factual, Non-Advisory Quantitative AI Boundary:** All price range projections, pullback probabilities, and sentiment scores are mathematical and historical outputs displayed beneath a static, non-flashy factual disclaimer header: *"Data & Quantitative AI Analysis Only — Not a Buy/Sell Recommendation."*
- **Custom Python Calculation Dominance:** External free APIs are used strictly via scheduled background jobs and a single multiplexed WebSocket worker to ingest raw primitives (OHLCV prices, 1-minute option candles, raw accounting lines, daily NAVs, and exchange filings). All financial ratios, multi-year CAGRs, Black-Scholes Option Greeks, valuation models, rolling returns, and probability bands are computed inside the platform's own Python Polars, NumPy, and SciPy backend services.
- **AI IDE & Coding Agent Architectural Flexibility Mandate:** The Modular Monolith folder structure, file organization, and SQL table definitions in this specification act as an authoritative reference architecture—not a rigid, unchangeable constraint. Any AI IDE, autonomous coding agent, or engineer implementing this system is explicitly authorized to adapt, split, merge, rename, or refactor directory layouts, database columns, and helper utilities as needed to optimize code maintainability, dependency compatibility, and runtime performance, provided 100% of the functional features and mathematical outputs remain intact.

2. Complete Multi-Page Site Map & Navigation Architecture

The frontend is organized into fifteen isolated, route-level workspaces under a unified top navigation bar with global instant search, a persistent ₹10,00,000 Zero-KYC Virtual Paper-Trading Sandbox drawer, hover-to-decode inline educational tooltips, and an interactive mega-dropdown for the AI Quant Hub.

Route & Module Name	Primary Target Assets	Core Capabilities, Calculators & Differentiators
1. Global Command Landing `/`	Nifty 50, Sensex, GIFT Nifty, Indian ADRs, India VIX, Sector Heatmaps	Instant autocomplete search across Stocks, Indices, F&O, IPOs, Mutual Funds, ETFs, and Bonds; live market breadth; top FII/DII flows; daily market mood summary.
2. Indices, GIFT Nifty, ADRs & India VIX Hub `/indices` & `/indices/[slug]`	All NSE/BSE Benchmark, Broad, Sectoral & Thematic Indices + GIFT Nifty + US-Listed Indian ADRs + India VIX	Live GIFT Nifty & Overnight Indian ADR (HDFC, ICICI, Infosys, Wipro, Dr. Reddy's) 9:15 AM Implied Opening Gap calculator, India VIX Market Stability & Expected Range calculator, Index Point Contributors, Constituent Weightages, 5Y Index P/E & P/B bands, Internal DMA Breadth, and direct Index F&O tabs.
3. Equities Deep-Dive `/stocks/[symbol]`	All NSE & BSE Listed Companies (5Y Annual + 20 Quarters + TTM)	Standalone/Consolidated toggle, Common-Size (%) toggle, Ind AS 108 segment split, Sector operational KPIs, 3-Pillar Health Score (Piotroski, Altman, Beneish), Volume Profile (POC) & Anchored VWAP chart overlays, DCF/Reverse-DCF calculator, Stock SIP calculator, Stock F&O tab, and On-Demand **+ Compare with Peers** drawer.

Route & Module Name	Primary Target Assets	Core Capabilities, Calculators & Differentiators
4. Futures & Options (F&O) & Option Chain Hub `/fno` & `/stocks/[symbol]/fno`	All Index Derivatives (Nifty, Sensex, Bank Nifty, FinNifty, Midcap, Bankex) & ~200 NSE F&O Stocks	Full Option Chain (Call/Put LTP, Call/Put OI & AOI, Volume, IV, Delta, Theta, Gamma, Vega, POP), Candlestick Call/Put OI Resistance/Support Overlays, Automated Tuesday (NSE) / Thursday (BSE) Expiry Calendar, Stock Physical Delivery Margin Trap Alerts, and **Mode A: Intraday & Multi-Day 1-Min Historical "What-If" Option Replay & Theta-Attribution Calculator**.
5. IPO Intelligence Hub `/ipos` & `/ipos/[slug]`	Mainboard & SME IPOs (Upcoming, Open, Closed, Listed)	3-to-5Y Restated DRHP Financials, Fresh Issue vs. OFS split, VC/Promoter Exit cost-vs-selling table, Pre-IPO Unlisted Share Price vs. IPO Price Band comparison, BRLM historical track record, live QIB/bNII/sNII/Retail subscription, GMP tracker, T+1 Direct Registrar Allotment link, 30/90-Day Anchor Lock-In alerts, and Post-Listing Monitoring Agency reports.
6. Mutual Funds Hub `/mutual-funds/[code]`	All AMFI Direct & Regular Equity, Debt & Hybrid Schemes	Rolling returns, Sharpe/Sortino/Ulcer ratios, Upside/Downside capture, Fund Manager tenure alpha, Monthly Portfolio Churn (New Entries & Exits) with Two-Tier AI Reason attribution, Backtested SIP/Lumpsum calculator, MF-vs-ETF True Exit Cost calculator, SWP & ₹1.25L Tax-Harvesting calculator, and On-Demand **+ Compare with Peers**.
7. ETFs Intelligence Hub `/etfs/[symbol]`	Equity, Gold, Silver, Debt & International ETFs on NSE/BSE	Live market price vs. real-time iNAV deviation, ETF Impact Cost, Tracking Error %, Liquidity scanner, and ETF vs. Index Mutual Fund net wealth comparison calculator.
8. Bonds & Debt Hub `/bonds`	RBI G-Secs, T-Bills, SDLs, Floating Rate Bonds, Corporate NCDs	Live India Sovereign Yield Curve (3M to 30Y), Credit Rating log, YTM & Yield-to-Call (YTC), Fixed vs. Floating Rate Bond calculator, and Clean Price vs. Dirty Price (Accrued Interest) post-tax calculator.
9. Commodities Hub `/commodities`	Gold & Silver (MCX Spot/Futures, 24K/22K Bullion, Listed SGBs)	5-Year Gold-to-Silver Ratio chart, Live 60+ SGB Secondary Market Discount/Premium scanner, and Physical Gold vs. Gold ETF vs. SGB comparative wealth calculator.
10. REITs & InvITs Hub `/reits-invits`	Listed Commercial Real Estate & Infrastructure Trusts	Mandatory 90% NDCF payout tracking, Occupancy %, WALE, Cap Rate, On-Demand Peer Compare, and 4-Component Distribution Post-Tax Yield calculator (Interest, Dividend, Debt Amortization, Other).
11. Corporate Events Hub `/events`	Dividends, Splits, Reverse Splits, Bonuses, Buybacks, Mergers, Demergers, Rights, Earnings	Full Ex-Date/Record-Date timeline calendar plus 5 dedicated calculators (Dividend & TDS, Split Adjustment, Bonus Tax-Stripping, Retail 15% Buyback Entitlement Arbitrage, Demerger Cost Split & Rights RE Fair Value) and ASM/GSM/F&O Ban logs.
12. 4 PM Market Pulse & Money Flow `/market-pulse`	NSE Sectoral Indices, Delivery Flows, FII/DII Cash & Derivatives	End-of-day money flow map separating Demat Delivery buying from intraday speculation; 5x Volume & Delivery Shockers; Sector Rotation across 1D/1W/1M horizons.

Route & Module Name	Primary Target Assets	Core Capabilities, Calculators & Differentiators
13. Portfolio X-Ray & Paper-Trading Sandbox `/portfolio-xray` & `/paper-trade`	Multi-Asset User Portfolios & ₹10,00,000 Virtual Sandbox	CAMS/CAS PDF & manual Portfolio X-Ray detecting cross-fund stock overlap and fee drag; Zero-KYC ₹10 Lakh Virtual Paper-Trading Sandbox across Shares, MFs, ETFs, and F&O with Net-in-Pocket statutory charge deductions.
14. AI Quant Hub `/ai-quant-hub`	4 Dedicated AI Workspaces (Hover Mega-Menu + Command Center)	(1) **RangeCast AI** (7D/14D Price Band & Pullback Engine), (2) **StreetPulse Matrix** (Big Firms vs. Retail Crowd), (3) **4 PM Market X-Ray**, and (4) **Ask Dalal AI** (Text-to-SQL, DRHP RAG, CEO Say-vs-Do Tracker).
15. Market Academy `/learn`	Universal **Fundamentals** + 7 Asset/F&O Tabs + 6 Forensic Guides	Structured educational encyclopedia starting with Universal Market Fundamentals, dedicated Indices/VIX/F&O Greeks masterclasses, asset-specific deep-dives, and universal Hover-to-Decode formula tooltips across every page.

3. Module-by-Module Functional & Mathematical Specifications

3.1 Module 1: Listed Equities, Technical Overlays & 5-Year Fundamental Engine (`/stocks/[symbol]`)

Every listed company page provides a comprehensive, institutional-grade breakdown across a 5-year annual window, a 20-quarter historical window, and a live Trailing Twelve Months (TTM) column. Users can switch globally between **Consolidated** (entire group including subsidiaries) and **Standalone** (parent entity only) statements, and activate a **Common-Size Toggle** that converts every Income Statement row into a percentage of Operating Revenue and every Balance Sheet row into a percentage of Total Assets.

Analysis Pillar	Exhaustive Metrics & Data Points Included	Backend Formula & Forensic Logic
Top-Level Quantitative Health Scorecard	Piotroski F-Score (0–9 scale), Altman Z-Score (Bankruptcy Risk), Beneish M-Score (Earnings Manipulation Detector).	Computed from 5Y balance sheet and cash flow deltas. Beneish M-Score greater than −1.78 flags potential accounting manipulation.
Market, Valuation Bands & Advanced Chart Overlays	Market Cap, CMP, 52W High/Low, Enterprise Value (EV), Stock P/E vs. Sector Median P/E, Book Value, P/B, EV/EBITDA, PEG Ratio, Graham Number, Dividend Yield %, 5-Year Historical P/E & P/B Chart with Median and ±1 SD bands. Interactive TradingView Candlestick chart with **Volume Profile (Point of Control — POC)**, **VWAP Anchored to Earnings/IPO/Expiry**, automated Candlestick Pattern detection, and 1-click **F&O Call/Put OI Support/Resistance Zones**.	*Graham Number* = $\sqrt{(22.5 \times EPS \times Book\ Value)}$. *EV = Market Cap + Total Debt − Cash & Liquid Investments*.

Analysis Pillar	Exhaustive Metrics & Data Points Included	Backend Formula & Forensic Logic
5Y & 20Q Income Statement + Ind AS 108 Segments	Operating Revenue (plus ARR for SaaS firms), Raw Material %, Employee Cost %, Finance Cost, D&A, EBITDA, OPM %, Effective Tax Rate %, PAT, EPS, 3Y/5Y Revenue & PAT CAGR. Full **Ind AS 108 Multi-Domain Segment Revenue, EBIT & Capital Employed** (e.g., Software Services vs. Hardware vs. Products) + Domestic vs. International Geographic split.	Automatically computes segment revenue share % and segment ROCE % to reveal which business vertical drives actual profitability vs. capital consumption.
5Y Balance Sheet, Working Capital & Hidden Risks	Share Capital, Reserves, Gross Debt, Net Debt, Fixed Assets, CWIP, Current Assets/Liabilities, Debt-to-Equity, Interest Coverage, Debtor Days, Inventory Days, Days Payable, Cash Conversion Cycle (CCC), Maintenance Capex vs. Growth Capex, R&D %, Contingent Liabilities, Related Party Transactions (RPT), Promoter Salary % of PAT, Auditor Resignations & CARO Flags, Unlisted Subsidiary Loans, 5Y Credit Rating History (CRISIL/ICRA/CARE) & Borrowing Cost %.	$CCC = Debtor\ Days + Inventory\ Days - Payable\ Days$. If Debtor Days rise faster than Revenue CAGR over 3 years, triggers an *"Aggressive Channel Stuffing / Uncollected Sales"* warning.
Cash Flow, Quality of Earnings & DuPont ROE	Cash from Operations (CFO), Investing (CFI), Financing (CFF), Free Cash Flow to Firm (FCFF), Free Cash Flow to Equity (FCFE), CFO-to-EBITDA %, CFO-to-PAT %, ROE %, ROCE %, ROA %, and 3-Step & 5-Step DuPont ROE decomposition.	$3\text{-}Step\ DuPont\ ROE = (PAT / Revenue) \times (Revenue / Total\ Assets) \times (Total\ Assets / Equity)$. Flags if cumulative 5Y CFO is less than 70% of cumulative 5Y PAT.
Industry-Specific Operational KPIs & Raw Materials	**Banking/NBFC:** NIM %, GNPA %, NNPA %, CASA %, Cost of Funds, PCR %, CRAR %. **IT Services:** Deal TCV (USD M), Attrition %, Utilization %, USD 1M+/50M+/100M+ Clients. **Healthcare:** ARPOB, Occupancy %, ALOS. **Telecom/Tech:** ARPU, Churn %, GOV/MTU. **Manufacturing/Infra:** Capacity Utilization %, Tonnage, EBITDA/Tonne, Order Book-to-Bill Ratio. **Retail:** SSSG %, Rev/Sq.Ft. **Supply Chain:** Raw Material Input Sensitivity & 5Y Product Market Share %.	Selects the exact operational KPI template dynamically based on the company's sector mapping so banks never show factory inventory metrics and manufacturers never show CASA ratios.
Smart Money, Shareholding & Insider Deals	Quarterly % held by Promoters, FIIs, DIIs (Mutual Funds & Insurance), and Retail; Promoter Pledging %; Monthly Mutual Fund Scheme-level buyers/sellers; NSE/BSE Bulk & Block Deals (>0.5% equity); SEBI SAST Insider Trading disclosures.	Cross-checks retail shareholder headcount growth against FII/DII stake changes to identify institutional distribution vs. accumulation.

Built-In Equity Calculators, Net-in-Pocket Tax Engine & On-Demand Peer Workspace:

1. **Interactive DCF & Reverse-DCF Intrinsic Value Calculator:** Pre-populates the company's 5-year Free Cash Flow, Net Debt, and WACC. Users can adjust growth sliders to compute intrinsic value per share, or flip to **Reverse-DCF Mode** to calculate the exact 10-year implied annual FCF growth rate currently priced into the live market price.
2. **Corporate-Action Adjusted Stock SIP, Lumpsum & Net-in-Pocket Breakeven Calculator:** Computes historical and forward returns for monthly stock SIPs or lumpsum purchases, factoring in historical Stock Splits, Bonus share credits, reinvested Cash Dividends (DRIP), and exact statutory trading deductions (Delivery vs. Intraday STT, Exchange Turnover Charges, SEBI Fees, Stamp Duty, DP Charges, and 18% GST).
3. **On-Demand `+ Compare with Peers` Workspace & Relative Valuation Calculator:** Triggered only when the user clicks the **+ Compare with Peers** button. Supports three comparison modes: (a) Direct Sub-Industry Peers, (b) Ind AS 108 Segment-Level Peers (for multi-business conglomerates), and (c) Custom User-Selected Peers. Automatically badges the #1 `Category Leader` in each metric row, persists comparison state across sub-tabs once opened, and includes a **Peer Relative Price Calculator** showing what Stock A's share price would be if valued at Peer Stock B's P/E, P/B, or EV/EBITDA multiple.
4. **Dual-Mode Custom Formula & Natural Language Stock Screener:** Allows users to filter all NSE/BSE listed companies using either mathematical syntax (for example, `ROCE > 20 AND DebtToEquity < 0.3 AND PromoterPledge = 0 AND CFOToEBITDA > 80`) or conversational English/Hinglish prompts routed through `Ask Dalal AI`.

3.2 Module 2: All-India Indices, GIFT Nifty, Indian ADRs & India VIX Stability Hub (`/indices`)

The Indices Hub makes every benchmark, broad-market, sectoral, thematic, volatility, and global pre-market derivative index visible and analyzable from a unified command center, tracking daily opening prices, intraday highs/lows, 3:30 PM closing values, constituent point contributions, and multi-year valuation bands.

- **Complete Universe of Visible NSE & BSE Indices:**
  - **Broad & Benchmark Indices:** Nifty 50, BSE Sensex (30), Nifty Next 50, Nifty 100, Nifty 200, Nifty 500, Nifty Midcap 50/100/150 (plus Nifty Midcap Select), Nifty Smallcap 50/100/250, Nifty Microcap 250, Nifty Total Market, BSE 100, BSE 200, BSE 500, BSE MidCap, BSE SmallCap, and BSE AllCap.
  - **Sectoral & Thematic Indices:** Nifty Bank (Bank Nifty), Nifty IT, Nifty Financial Services (FinNifty), BSE Bankex, Nifty Auto, Nifty Pharma, Nifty FMCG, Nifty Metal, Nifty Realty, Nifty Energy, Nifty PSU Bank, Nifty Private Bank, Nifty Healthcare, Nifty Oil & Gas, Nifty Consumer Durables, Nifty Media, Nifty India Defence, Nifty EV & New Age Automotive, Nifty CPSE, Nifty PSE, Nifty Infrastructure, and Nifty Commodities.

- **GIFT Nifty + Overnight US-Listed Indian ADR 9:15 AM Opening Gap Engine:**
  - Tracks **GIFT Nifty (NSE IX — GIFT City, Gujarat)** across its nearly 21-hour trading window (Session 1: 6:30 AM – 3:40 PM IST; Session 2: 4:35 PM – 2:45 AM IST). Before the domestic market opens at 9:15 AM IST, the backend computes the **Implied Nifty 50 Spot Opening Gap (Points & %)** by comparing live GIFT Nifty Futures against Nifty 50's previous 3:30 PM close adjusted for fair-value Futures basis.
  - Simultaneously tracks overnight US-listed **Indian ADRs (American Depositary Receipts)** including `HDFC Bank (HDB)`, `ICICI Bank (IBN)`, `Infosys (INFY)`, `Wipro (WIT)`, `Dr. Reddy's (RDY)`, and `MakeMyTrip (MMYT)` alongside Nasdaq, S&P 500, Nikkei 225, Hang Seng, Brent Crude, and Dollar Index (DXY) so users know at 8:30 AM which specific Indian sector (Banking, IT, or Pharma) is poised to lead or drag the market at 9:15 AM.
- **India VIX (Volatility Index) Market Stability & Expected Range Engine:**
  - Although retail traders cannot directly buy or sell India VIX contracts, India VIX is the primary gauge of market stability and option pricing fear. The platform displays live and historical India VIX levels, classifies current market stability into `Low-Volatility / Stable (<13)`, `Normal Regime (13–18)`, or `High-Fear / Turbulent (>18)`, and automatically calculates the mathematically implied Nifty 50 movement range:

    ```
    Expected Daily Swing (%)   = ± (India VIX / √252)
    Expected Weekly Swing (%)  = ± (India VIX / √52)
    Expected Monthly Swing (%) = ± (India VIX / √12)
    ```

  - Includes an automated **Pre-Event IV Crush Warning** ahead of Union Budget days, RBI MPC policy announcements, Election results, and quarterly earnings, warning option buyers when inflated VIX is likely to collapse and crush both Call and Put premiums simultaneously.
- **Inside Every Individual Index Page (`/indices/[slug]`):** Displays (1) **Index Point Contributors** showing exact points added or dragged down by each constituent stock today, (2) **Full Constituent & Free-Float Weightage Table**, (3) **5-Year Historical Index P/E, P/B & Dividend Yield Valuation Bands** (with Median, +1/+2 SD overbought, and −1/−2 SD oversold lines), and (4) **Internal DMA Breadth Meter** (% of index stocks trading above their 20-DMA, 50-DMA, and 200-DMA).

3.3 Module 3: Stock & Index Futures & Options (F&O), Candlestick OI Overlays & Mode-A Historical Option Replay Calculator (`/fno` & `/stocks/[symbol]/fno`)

Every derivative-enabled Index (`Nifty 50`, `BSE Sensex`, `Bank Nifty`, `FinNifty`, `Nifty Midcap Select`, `Nifty Next 50`, `BSE Bankex`) and all ~200 NSE F&O-listed individual companies (such as Reliance, HDFC Bank, Policybazaar/PB Fintech, Tata Motors, Infosys, etc.) feature a dedicated, full-stack **F&O & Option Chain Workspace**. If a user opens a cash-only company that is not part of the F&O segment, the tab displays a contextual card: `[Cash Segment Only — Not in SEBI F&O List]` along with a live tracker showing where the stock stands relative to SEBI's quantitative F&O eligibility rules (Market-Wide Position Limit, Median Quarter Sigma Order Size, and Average Daily Delivery Value).

F&O Component	Metrics, Greeks & Visual Overlays	Backend Quantitative & SEBI Rule Logic
Complete Live & EOD Option Chain Matrix	For every Call (CE) and Put (PE) strike across all active expiries: **Call Price (LTP)**, **Put Price (LTP)**, **Total Call OI**, **Total Put OI**, **Call Change in OI (AOI)**, **Put Change in OI (AOI)**, Volume, Bid/Ask Spread, Strike PCR, **Implied Volatility (IV %)**, IV Rank (IVR), IV Percentile (IVP), **Delta (Δ)**, **Gamma (Γ)**, **Theta (Θ — Daily Time Decay in ₹)**, **Vega (v)**, and **Probability of Profit (POP %)**.	Computes IV via Newton-Raphson root-finding and computes all five Black-Scholes-Merton Greeks and POP % natively inside FastAPI using `NumPy` and `SciPy` at zero vendor cost: $POP\ (Long\ Call) = N(d_2)$ at breakeven strike.
Direct Candlestick Chart F&O Smart Overlays	Toggleable directly on the 5m, 15m, 1H, and Daily TradingView Candlestick Chart: (1) **Red Shaded Resistance Zone** at the Strike with Maximum Call OI & Maximum Intraday Call AOI, (2) **Green Shaded Support Zone** at the Strike with Maximum Put OI & Maximum Intraday Put AOI, (3) **Live Max Pain Strike Line**, and (4) **Synchronized Sub-Chart OI & AOI Histogram** beneath Volume bars.	Lets users visually verify whether a bullish or bearish candle breakout is supported by real option-writer unwinding (Short Covering) or blocked by a massive Call/Put OI wall.
Automated NSE & BSE Expiry Calendar Engine	Automatically manages exchange-specific weekly and monthly expiry dates and holiday shifts: **Nifty 50 Weekly & Monthly Expiry (Every Tuesday)**, **BSE Sensex Weekly & Monthly Expiry (Every Thursday)**, **Bank Nifty / FinNifty / Midcap Select / All Individual Stock F&O Monthly Expiry (Last Tuesday of the Month)**, and **BSE Bankex Monthly Expiry (Last Thursday of the Month)**.	If a scheduled Tuesday or Thursday expiry falls on an NSE/BSE trading holiday, the calendar engine automatically shifts the effective expiry timestamp to the immediately preceding trading session.
Futures 4-Quadrant Buildup, Rollover & Physical Settlement Alerts	Near/Next/Far Month Futures price, Spot-to-Futures Basis (Premium/Discount), Cost of Carry %, 3M/6M Rollover %, **Long Buildup (Price ↑, OI ↑)**, **Short Buildup (Price ↓, OI ↑)**, **Short Covering (Price ↑, OI ↓)**, **Long Unwinding (Price ↓, OI ↓)**, live **MWPL % (95% F&O Ban Meter)**, and **Participant-Wise OI (FII vs. DII vs. Pro vs. Retail Client)**.	Includes an automated **Stock F&O Physical Settlement & Margin Trap Alert** starting 4 trading days before the Last Tuesday monthly expiry for all In-The-Money (ITM) individual stock options and futures.

> **Flagship Derivatives Calculator — Mode A: Intraday & Multi-Day 1-Minute Historical "What-If" Option Replay & Theta-Attribution Calculator:**
> Many Indian platforms show only static option chains without letting a user inspect exact historical intraday option price behavior and time-decay erosion between two specific timestamps. The platform implements **Mode A (Historical Intraday & Multi-Day Timestamp Replay Engine)**:

- **User Inputs:** The user selects any Index or Stock Option strike (for example, `Sensex 70,400 PE` or `Nifty 24,500 CE` or `Reliance 1,500 CE`), selects a specific **Entry Date & Timestamp** (e.g., `10:00 AM`), selects a specific **Exit Date & Timestamp** (e.g., `10:10 AM`, `10:30 AM`, `2:15 PM`, or next day `9:30 AM`), and inputs their capital invested or number of lots.
- **Exact 1-Minute Historical Price & Net-in-Pocket Output:** The backend queries the stored 1-minute option and underlying spot hypertables (`TimescaleDB`) and displays the exact factual replay:

```
Selected Contract : Sensex 70,400 PE (Thursday Expiry)
Entry Timestamp   : 10:00 AM  | Option Price: ₹185.00 | Underlying Sensex
Spot: 70,480
Exit Timestamp    : 10:10 AM  | Option Price: ₹212.00 | Underlying Sensex
Spot: 70,390
Price Change      : +₹27.00 per unit (+14.59% Gain in 10 Minutes)
Gross Value       : ₹50,000 Invested (270 Qty) → ₹57,297 Gross Value (+
₹7,297 Gross Profit)
Statutory Charges : −₹157 (Brokerage + Option Sell STT + Exchange Turnover
+ GST + Stamp Duty)
Net-in-Pocket P&L : +₹7,140 Net Profit (+14.28% Net Return)
```

- **Exact Greek & Theta-Decay Attribution Breakdown:** Directly below the P&L output, the Python Black-Scholes attribution engine decomposes the exact ₹27.00 option price change between `10:00 AM` and `10:10 AM` (or across any multi-hour/multi-day window) so the user explicitly sees how much money was gained from the market move versus how much was eaten by **Theta (Time Decay)**:

```
1. Underlying Spot Move Impact (Delta + Gamma) : +₹26.15 (+14.13%)
2. Implied Volatility Shift Impact (Vega)      : +₹1.45  (+0.78%)
3. Time Decay Erosion During Window (Theta)    : −₹0.60  (−0.32% lost
purely to time passage)
--------------------------------------------------------------------------
-----------------
Net Option Price Change (10:00 AM → 10:10 AM)  : +₹27.00 (+14.59%)
```

3.4 Module 4: IPO Intelligence Hub — Mainboard & SME (`/ipos`)

The IPO Hub provides full transparency into every Mainboard and SME public issue so retail, Small HNI (sNII), Big HNI (bNII), and QIB investors can inspect the underlying fundamentals and insider exit math rather than relying blindly on speculative chatter.

- **3-to-5-Year Restated Pre-IPO Financial Statements (DRHP/RHP Deep-Dive):** Extracts the complete 3-to-5-year Restated Income Statement, Balance Sheet, Cash Flow, Net Worth, Borrowings, and Return Ratios directly from the SEBI DRHP/RHP filings so users can analyze unlisted IPO companies with the exact same depth as listed equities.

- **Objects of the Issue & OFS Investor Exit Forensics:** Breaks down total issue size into **Fresh Issue** (capital entering the company for Debt Repayment, Capex, or Working Capital) versus **Offer For Sale / OFS** (early Venture Capital funds and promoters exiting). A dedicated **Selling Shareholder Table** lists every exiting entity, their weighted-average pre-IPO acquisition cost per share, the IPO offer price, their exact return multiple (e.g., 42x return), and their post-IPO retained holding percentage.
- **Pre-IPO Unlisted Market Price vs. IPO Price Band Tracker:** Tracks historical prices of the company's shares in the unofficial **Pre-IPO Unlisted Market** alongside Pre-IPO private placement rounds executed within 6 months prior to the RHP, revealing whether the official IPO Price Band is priced at a fair discount or an aggressive markup relative to unlisted trading history.
- **Merchant Banker (BRLM) Track Record & SME Liquidity Audit:** Displays the historical track record of the Book Running Lead Managers (BRLMs), calculating the average listing gain and 1-year post-listing survival rate across their last 10 managed IPOs. For **SME IPOs**, explicitly tracks the designated **Market Maker** name, mandatory Market Maker inventory quota, minimum lot application value (₹1 Lakh+), and eligibility criteria for future migration from BSE SME / NSE Emerge to the Mainboard.
- **Live Subscription, GMP & Direct T+1 Registrar Allotment Engine:** Tracks live day-by-day subscription multipliers across QIB, bNII, sNII, Retail, and Employee quotas alongside Grey Market Premium (GMP) labeled as an unofficial market indicator. Each IPO stores its official registrar mapping (`KFintech`, `MUFG Intime / Link Intime`, `Bigshare`, `Purva Sharegistry`, `Skyline`, `Cameo`, or `Maashitla`). Once bidding closes, a prominent **Check Live Allotment Status** button routes the user directly to the official registrar portal on T+1 allotment day, accompanied by mathematical allotment probability odds parsed from the Basis of Allotment filing.
- **Anchor Lock-In Expiry Alerts & Post-Listing Proceeds Monitoring:** Automatically schedules calendar alerts for the **30-Day Anchor Lock-In Expiry** (50% of anchor shares unlock) and **90-Day Anchor Lock-In Expiry** (remaining 50% unlock), and ingests quarterly Monitoring Agency Reports for ₹100 Crore+ issues to verify actual utilization of Fresh Issue funds.

3.5 Module 5: Mutual Funds & ETFs Intelligence Hub (`/mutual-funds & /etfs`)

Every Mutual Fund and ETF page provides deep quantitative performance metrics, complete monthly portfolio holdings, automated tracking of stock additions and exits with AI-verified reasons, and a four-part calculator suite.

- **Quantitative Risk, Capture & Manager Alpha Metrics:** Computes 1Y, 3Y, 5Y, and 10Y Rolling Returns, Alpha, Beta, Sharpe Ratio, Sortino Ratio, Standard Deviation, Maximum Drawdown, **Upside Capture Ratio**, **Downside Capture Ratio**, **Ulcer Index**, **Portfolio Turnover Ratio (%)**, and **Active Share (%)**. Separates historical returns into performance achieved under the current Fund Manager's tenure versus prior managers. For Debt and Hybrid funds, displays Yield to Maturity (YTM %), Modified Duration, Average Maturity, and Credit Rating Quality Breakdown (Sovereign, AAA, AA, and Below-AA downgrade warnings).

- **Monthly/Quarterly Portfolio Churn Tracker & Two-Tier AI Reason Engine:** By comparing monthly AMFI portfolio disclosures (Month *T* vs. Month *T−1* and Quarter *Q* vs. Quarter *Q−1*), the backend categorizes stock movements into **Fresh Additions (100% New Entry)**, **Complete Exits (100% Sold Off)**, **Stake Increased**, and **Stake Trimmed**. To explain *why* a fund house added or dumped a stock without hallucinating:
  1. **Tier 1 — Official AMC Commentary Badge (`[Official AMC Reason]`):** Extracts verbatim explanations from the monthly Fund Manager Factsheet commentary if provided by the fund house.
  2. **Tier 2 — AI Verified Market Event Correlation (`[AI Verified Market Event]`):** If the AMC did not state a reason, the backend AI pipeline queries verified exchange filings, CRISIL/ICRA credit rating actions, NCLT bankruptcy logs, SEBI orders, and quarterly earnings releases from that exact month. If a verified event occurred (such as a credit downgrade to Default, bankruptcy filing, or 60% earnings drop), it displays the verified event with clickable source citations; otherwise, it displays `[Reason Not Provided by AMC — Standard Valuation/Weightage Rebalancing]`.
- **4-in-1 Mutual Fund & ETF Interactive Calculator Suite:**
  1. **Historical Backtested SIP & Lumpsum Calculator:** Connected directly to daily historical NAV series to compute exact historical XIRR and maturity value alongside forward CAGR projections.
  2. **Mutual Fund vs. Equivalent ETF "True-Cost & Exit" Calculator:** Whenever a user opens an Index, Sectoral, or Commodity Mutual Fund that has a matching listed ETF, the platform highlights the equivalent ETF and calculates the exact rupee difference at redemption. The backend formula incorporates every real-world friction cost:

     ```
     MF Net Maturity  = Gross Corpus − (Daily Compounded MF TER + Exit Load % on
     Redemption + 0.005% Stamp Duty)
     ETF Net Maturity = Gross Corpus − (Daily Compounded ETF TER + Brokerage +
     0.001% Delivery Sell STT + ₹15 DP Charge + iNAV Impact Cost %)
     ```

  3. **On-Demand `+ Compare with Peers` Drawer & Relative Wealth / Fee-Drag Calculator:** Triggered via the **+ Compare with Peers** button. Benchmarks the scheme against all Direct-Growth schemes in the same SEBI category, displays a **Portfolio Stock Overlap Matrix (%)**, and calculates the exact rupee Alpha Difference and Fee-Drag Difference between Fund A and Peer Fund B.
  4. **SWP & ₹1.25 Lakh LTCG Tax-Harvesting Calculator:** Models monthly Systematic Withdrawal Plans and computes the exact number of units a user can redeem and repurchase before March 31 each year to harvest India's ₹1,25,000 tax-free Long-Term Capital Gains exemption.

3.6 Module 6: Bonds, Commodities (Gold/Silver), REITs/InvITs & Virtual Paper-Trading Sandbox

- **Government & Corporate Bonds Hub (`/bonds`):** Covers RBI Retail Direct instruments (G-Secs, 91D/182D/364D T-Bills, State Development Loans, Floating Rate Savings Bonds) and listed Corporate Bonds/NCDs. Features the live **India Sovereign Yield Curve** (3M to 30Y), credit rating logs, **Fixed vs. Floating Rate Bond Calculator** (modeling NSC/T-Bill benchmark resets), and a **Clean Price vs. Dirty Price & Post-Tax YTM/YTC Calculator** that accounts for accrued interest since the last coupon payment date.
- **Gold & Silver Commodities Hub (`/commodities`):** Tracks live MCX Spot and Futures contracts, IBJA 24K/22K bullion rates, and the 5-Year Gold-to-Silver Ratio. Includes a **Live Listed SGB Secondary Market Scanner** (ranking all 60+ listed Sovereign Gold Bond tranches by discount/premium to fair value plus remaining 2.5% coupons) and a **Physical Gold vs. Gold ETF vs. SGB Comparative Calculator**.
- **REITs & InvITs Hub (`/reits-invits`):** Tracks Embassy, Mindspace, Brookfield, Nexus Select, PowerGrid InvIT, and IndiGrid across Occupancy %, WALE, Cap Rate, NDCF payout compliance, On-Demand Peer Comparison, and a **4-Component Post-Tax Distribution Calculator** (breaking payouts into Interest, Dividend, Tax-Free Debt Amortization, and Other Income).
- **Zero-KYC ₹10,00,000 Virtual Paper-Trading & Strategy Sandbox (`/paper-trade`):** Gives every visitor (including recruiters and hiring managers testing the live platform) an instant ₹10 Lakh virtual balance to simulate buying Equities, Mutual Fund SIPs, ETFs, and F&O Option strategies at live prices, tracking real-time mark-to-market P&L after statutory charges.

3.7 Module 7: Dedicated Corporate Events & Market Calendar Hub (`/events`)

A dedicated top-level navigation page tracks every corporate and macro event across four lifecycle timestamps (**Announcement Date → Ex-Date → Record Date → Payout/Effective Date**) paired with five specialized event calculators:

Event Category	Tracked Data Points & Lifecycle Dates	Embedded Event Calculator
Dividends (Interim, Final, Special)	Dividend per Share (₹), Face Value %, Effective Yield % at CMP, Ex-Date, Record Date, Payout Date.	**Dividend & TDS Post-Tax Calculator:** Computes gross cash payout, 10% TDS deduction threshold (>₹5,000/year), and net post-tax yield.
Stock Splits & Reverse Splits	Old Face Value vs. New Face Value, Split/Consolidation Ratio, Ex-Split Date, Record Date.	**Split Adjustment Calculator:** Computes new share count, adjusted average buy price, and fractional share cash payout.
Bonus Issues	Bonus Ratio (e.g., 1:1, 1:2), Ex-Bonus Date, Record Date, T+2 Share Credit Date.	**Bonus & FIFO Tax-Stripping Calculator:** Computes post-bonus price and splits cost of acquisition (original shares vs. ₹0-cost bonus shares).

Event Category	Tracked Data Points & Lifecycle Dates	Embedded Event Calculator
Share Buybacks (Tender vs. Open Market)	Issue Size (₹ Cr), Buyback Price, Premium % over CMP, Record Date, Tender Window Open/Close, Payout Date.	**Retail 15% Quota Buyback Arbitrage Calculator:** Computes maximum shares for the ₹2,00,000 Small Shareholder quota, Acceptance Ratio %, and net arbitrage profit.
Mergers, Demergers & Rights Issues	NCLT status, Share Swap Ratio, Demerged Entity listing timeline, Rights Ratio/Discount %, and Rights Entitlement (RE) trading dates.	**Demerger Cost-Split & Rights Entitlement (RE) Calculator:** Splits original purchase price using official Net Book Value ratios for demergers, and calculates theoretical Ex-Rights price and fair value of tradable REs.
Earnings, Lock-Ins, Rebalancing & Surveillance	Q1–Q4 Result dates & Beat/Miss badges, Listed OFS/QIP windows, 30D/90D/6M/1Y Post-IPO Lock-In Expiries, Nifty/MSCI Rebalancing flows, SEBI ASM/GSM & NSE F&O Ban lists, RBI MPC & Macro dates.	**Passive Flow & Free-Float Shock Estimator:** Estimates million-share supply unlocked on lock-in expiry or passive ETF inflows on index inclusion.

3.8 Module 8: Top-Navbar `AI Quant Hub` (`/ai-quant-hub`) — 4 Branded Workspaces

Placed prominently in the top navigation bar, hovering over **AI Quant Hub** reveals an instant 4-card mega-dropdown, while clicking it opens the unified command center page. Every workspace renders beneath the non-advisory factual banner (*"Data & Quantitative AI Analysis Only — Not a Buy/Sell Recommendation"*).

1. **Workspace 1 — `RangeCast AI` (7-Day & 14-Day Probabilistic Price Band & Pullback Engine):**
   - Combines three factual quantitative signals: (a) **NSE F&O Open Interest (OI) Order Walls** identifying Long Buildup, Short Buildup, Short Covering, Long Unwinding, Maximum Call OI ceiling, and Maximum Put OI floor; (b) **Delivery Volume Buy/Sell Pressure Z-Score**; and (c) a **Quantile Time-Series Model (`LightGBM`)** with 5-year historical regime backtesting.
   - Outputs the mathematical 7-Day and 14-Day expected trading range (80% confidence interval) and exact historical pullback frequency (for example, *"Stock is trading +2.7 standard deviations above its 50-DMA; across 14 identical occurrences in the last 5 years, the stock consolidated or pulled back by 4.1% to 6.5% within 10 trading days in 78.5% of instances"*).
2. **Workspace 2 — `StreetPulse Matrix` (Big Institutional Sharks vs. Retail Crowd Consensus):**
   - **What Big Firms Are Doing:** FII & DII Cash, Index Futures, Index Options, and Stock Futures net flows; Mutual Fund and Company Insider net accumulation; and aggregated public research targets from Tier-1 institutional brokerages (Highest, Average, and Lowest price target plus core rationale).

3. ◦ **What Small Firms & Retail Crowd Are Thinking:** NSE Retail delivery participation, Margin Trading Facility (MTF) leveraged buying book changes, and local `ProsusAI/finbert` sentiment classification across Indian financial Reddit communities, FinTwit, and RSS news feeds. Automatically flags **Social Hype vs. Fundamentals Divergences** when retail sentiment exceeds 85% bullish while insiders/FIIs are selling.
4. **Workspace 3 — `4 PM Market X-Ray` (Automated End-of-Day Money Flow & Sector Rotation):**
   - Generated automatically at 4:00 PM IST after market close. Ranks all NSE sectors across 1-Day, 1-Week, and 1-Month net institutional inflows, separates genuine Demat Delivery buying from intraday speculation, flags 5x Volume & Delivery Shockers (*Quiet Institutional Accumulation* vs. *Speculative Retail Bull Traps*), and maps global macro movements (Brent Crude, USD/INR, US 10Y Yield, GIFT Nifty, Indian ADRs, India VIX) to impacted domestic sectors.
5. **Workspace 4 — `Ask Dalal AI` (Multi-Agent Deep Research Copilot, CEO Credibility Tracker & Semantic Diff):**
   - **Agentic Text-to-SQL Screener:** Translates natural language queries in English or Hinglish into safe, read-only PostgreSQL queries over pre-computed views.
   - **DRHP & Annual Report Hybrid RAG:** Parses complex financial tables from SEBI DRHPs and Annual Reports using `Docling` and executes hybrid vector + keyword retrieval via `pgvector`.
   - **The "CEO Say-vs-Do" Credibility Tracker:** Extracts forward-looking guidance statements from the past 3 years of quarterly Concall transcripts and verifies them against actual reported PostgreSQL financial numbers in subsequent years to compute an objective **Management Execution Score**.
   - **Quarterly Filing Semantic Diff & Event Reaction Backtester:** Highlights newly added risk warnings between sequential quarterly filings and backtests 1-day, 5-day, and 15-day stock return distributions across the last 12 quarterly earnings announcements and RBI policy dates.

3.9 Module 9: Interactive `Market Academy (/learn)` & Hover-to-Decode Tooltips

- **Part A — Dedicated Top-Navbar `Market Academy (/learn)`:**
  - **Top Featured Section — `Fundamentals (Universal Market Basics)`:** Covers foundational concepts common to all investors before entering asset-specific tabs: NSE/BSE mechanics, Role of SEBI, Demat vs. Trading accounts, T+1 settlement, Market vs. Limit orders, Upper/Lower circuits, Simple vs. Compounding math, CAGR vs. XIRR, Inflation & Real Return calculator, Indian Capital Gains Taxation (STCG/LTCG across Equity, Debt, Gold, and REITs), **Statutory Trading Costs (STT, Stamp Duty, Exchange Turnover Fees, SEBI Charges, GST, and Net Breakeven Math)**, **Global Pre-Market Cues (GIFT Nifty, Indian ADRs, Dollar Index DXY, US Bond Yields)**, RBI Repo Rate, and CPI Inflation.

- ◦ **7 Asset & Derivatives Learning Tabs:**
  1. **Shares (Equities & Technical Overlays):** P/E, P/B, EV/EBITDA, ROE, ROCE, DuPont Analysis, Free Cash Flow, Working Capital Days, Promoter Pledging, Bulk/Block Deals, Delivery %, **Volume Profile (Point of Control — POC)**, and **Anchored VWAP**.
  2. **Indices, India VIX & F&O (Derivatives Masterclass):** Free-Float Index Weightage math, SEBI's **Tuesday (Nifty) & Thursday (Sensex) Expiry Rules**, **India VIX Stability & IV Crush Math**, **Option Chain Decoding (Call/Put LTP, Total OI, Change in OI [AOI], IV, IVR, IVP)**, interactive visual guides for the **5 Option Greeks (Delta, Gamma, Theta Decay, Vega, POP %)**, Max Pain, the 4 Futures Buildup Quadrants, MWPL 95% Ban Rule, and **Cash vs. Compulsory Physical Delivery Settlement**.
  3. **IPOs (Mainboard, SME & Unlisted Market):** DRHP vs. RHP, Fresh Issue vs. OFS Investor Exits, **Pre-IPO Unlisted Share Price vs. IPO Price Band**, BRLM Track Record, QIB/bNII/sNII/Retail Quotas, Anchor 30/90-Day Lock-In, Registrar Basis of Allotment, and GMP.
  4. **Mutual Funds & ETFs:** NAV vs. iNAV, Expense Ratio (TER), Exit Load, Direct vs. Regular, ETF Impact Cost, Tracking Error, Upside/Downside Capture, Sharpe/Sortino/Ulcer Ratios, Rolling Returns, and Portfolio Overlap.
  5. **Bonds (Govt & Private):** Face Value, Coupon Rate, YTM vs. Yield-to-Call (YTC), Clean Price vs. Dirty Price (Accrued Interest), G-Secs, T-Bills, SDLs, Floating vs. Fixed Rate Bonds, and Credit Ratings.
  6. **Gold & Silver (Commodities):** MCX Spot vs. Futures, 24K vs. 22K Purity, Listed SGB Secondary Market Discounts vs. Gold ETFs vs. Physical Gold, and Gold-Silver Ratio.
  7. **REITs & InvITs:** Mandatory 90% NDCF distribution rules, 4-Component Distribution Tax Split, WALE, and Cap Rates.
- ◦ **6 Advanced Practical & Forensic Modules:** (1) Financial Forensics & Accounting Tricks, (2) Behavioral Finance & Operator Pump-and-Dump Traps, (3) Corporate Actions & Price Adjustment Math, (4) Multi-Leg Option Hedging & Payoff Mechanics, (5) Portfolio Construction & Asymmetric Drawdown Math, and (6) Indian Regulatory Protection & Filing SEBI SCORES Complaints.
- **Part B — Universal "Hover-to-Decode" Inline Tooltips:** Every financial metric and Greek label across the entire web application (such as `ROCE`, `AOI`, `Theta`, `IVP`, `OFS`, or `Dirty Price`) carries a subtle interactive underline. Hovering over any term opens a micro-card displaying: (1) Plain-English & Hinglish-friendly 1-line meaning, (2) Exact mathematical formula used by the backend, (3) Sector or volatility rule-of-thumb benchmark, and (4) A direct link to the full visual lesson inside `Market Academy`.

4. High-Level System Design (HLD) & Cloudflare Zero-Trust Topology

The system implements a **Modular Monolith Backend Architecture** paired with a **Cloudflare-Fronted Zero-Trust Edge Topology**. Pushing code to the GitHub repository triggers automated CI/CD deployment of the Next.js multi-page frontend to **Cloudflare Pages**, while the heavy Python FastAPI, Polars, FinBERT, and PostgreSQL (`TimescaleDB` + `pgvector`) stack runs inside Docker Compose connected exclusively via an outbound encrypted **Cloudflare Zero-Trust Tunnel (`cloudflared`)** with zero open inbound firewall ports.

```
[User Browser (Desktop & Mobile)]
       │
       ▼
[Cloudflare Global Edge Network (Custom Domain + Automated GitHub CI/CD)]
  ├── Cloudflare WAF + DDoS Protection + Redis Token-Bucket Rate Limiting
  ├── Cloudflare Pages ──────────► Serves Next.js App Router Multi-Page UI at Edge
  ├── Cloudflare Edge Cache ──► Caches 4 PM EOD Reports & 5Y Ratios (0ms Hit)
  ├── Cloudflare R2 Storage ──► Stores SEBI DRHP PDFs, XBRLs & Annual Reports (₹0
Egress)
  └── Cloudflare Zero-Trust Tunnel (cloudflared — Outbound Encrypted Private Link)
            │
            ▼ (Zero Public IP Exposure / Zero Open Inbound Ports)
[Private Backend Compute Node (Docker Compose Production Stack)]
  ├── FastAPI Modular Monolith (Async REST + Redis Pub/Sub WebSocket Multiplexer)
  ├── Quantitative Engine (Polars, NumPy, SciPy Black-Scholes Greeks, LightGBM)
  ├── Local AI & Failover Router (ProsusAI/FinBERT, Docling, Multi-LLM Router)
  ├── Redis 7 (Tier-2 Distributed Cache, Celery Broker & Dead Letter Queue)
  └── PostgreSQL 16 (Single Instance with TimescaleDB + pgvector Extensions)
```

Key Backend System Design Pillars for Engineering Interviews

1. **Multi-Tier Read-Through Caching:** Tier 1 (FastAPI in-memory LRU cache for symbol/ISIN/index metadata, 0ms), Tier 2 (Redis 7 distributed cache for computed 5Y ratios, option chains, and EOD flows, *<5ms*), and Tier 3 (PostgreSQL Materialized Views refreshed post-market).
2. **CQRS-Lite & Direct XBRL Ingestion:** Separates heavy background Write workers (parsing official BSE/NSE XBRL accounting tags and EOD Bhavcopies) from fast Read endpoints serving pre-computed JSON views.
3. **Corporate-Action Price & Option Back-Adjustment Pipeline:** Stores both `rawClose` and `adjClose` (via cumulative `adjustmentFactor`) in `TimescaleDB` so Stock Splits, Bonus issues, and Rights issues never distort historical charts, moving averages, or `RangeCast AI` calculations.
4. **Single-Connection WebSocket Multiplexer (Redis Pub/Sub Fan-Out):** Because free broker APIs allow only 1 to 3 concurrent WebSocket connections per API key, a single background Python ingestion worker maintains one WebSocket connection to the broker feed, publishes live index, stock, and option ticks to **Redis Pub/Sub**, and fans them out to thousands of concurrent browser clients via FastAPI WebSockets.

5. **Circuit Breakers & Dead Letter Queues (DLQ):** Every external data fetcher is wrapped in a Circuit Breaker; if fetching 1 symbol out of 2,000 fails, that job is routed to a Redis Dead Letter Queue with exponential backoff so the remaining 1,999 symbols complete without interruption.

5. 100% Zero-Budget (₹0) Data Ingestion & Multi-LLM Failover Strategy

Data / AI Layer	Verified Free Source / Provider	Ingestion Mode & Zero-Cost Guarantee
NSE/BSE Equities, Indices, VIX & XBRL Financials	`nsepython`, `jugaad-data`, `yfinance`, NSE/BSE Public XBRL & EOD Bhavcopy Archives	**₹0 Forever:** Pulled via scheduled Celery cron workers after 3:30 PM IST close into PostgreSQL.
Live 1-Min Candles, Indices & Full F&O Option Chains	`Upstox Developer API` / `ICICIdirect Breeze API` / `Angel One SmartAPI` + NSE F&O Bhavcopy & Participant OI CSVs	**₹0 API Subscription:** Multiplexed via single WebSocket into Redis Pub/Sub and stored in 1-minute `TimescaleDB` hypertables for Mode-A Replay.
GIFT Nifty & Overnight US-Listed Indian ADRs	NSE International Exchange (NSE IX) Public Feed + Yahoo Finance Global Tickers (`HDB`, `IBN`, `INFY`, `WIT`, `RDY`)	**₹0 Forever:** Polled every 5 minutes during pre-market hours (6:30 AM – 9:15 AM IST) to compute the Implied Opening Gap.
Mutual Funds (10,000+ Schemes) & Monthly Holdings	`MFapi.in` (`https://api.mfapi.in`) + AMFI India Public NAV & Monthly Portfolio Disclosures	**₹0 Forever:** No API key required; synced daily for NAVs and monthly for portfolio churn detection.
IPOs, Unlisted Prices, Registrars, Bonds & Bullion	NSE/BSE IPO & Debt JSON Feeds, SEBI DRHP RSS, RBI DBIE Portal, IBJA Bullion, Playwright Scraper	**₹0 Forever:** Runs twice daily (1:00 PM & 6:00 PM IST) behind Circuit Breakers.
Zero-Cost Multi-LLM Failover Router & Local CPU Models	**Primary:** Google AI Studio (`Gemini 2.5 Flash` & `Gemini Embedding` Free Tier) **Failover 1:** Groq Cloud Free Tier (`Qwen` / `Llama` LPU) **Failover 2:** OpenRouter (`DeepSeek R1/V3 :free`) **Local CPU:** `ProsusAI/finbert`, `Docling`, `LightGBM`	**₹0 Forever (No Credit Card Required):** Automatic *<15ms* failover on HTTP 429 rate limits, backed by 24-hour semantic response caching in Redis.

6. Reference Modular Monolith Repository Structure

Note for AI IDE / Coding Agent: This directory layout is a clean reference blueprint. You may add, rename, or reorganize sub-packages and files as needed during implementation.

```
bharat-fintech-intelligence/
├── docker-compose.yml                  # PostgreSQL 16 (TimescaleDB + pgvector),
Redis 7, FastAPI, Celery
├── .github/workflows/ci-cd.yml         # Automated pytest + Cloudflare Pages &
Tunnel deployment
├── backend/
│   ├── app/
│   │   ├── main.py                     # FastAPI entry point, CORS, WebSocket
Multiplexer mount
│   │   ├── core/                       # Config, Redis Cache, DB Async Session,
Token-Bucket Limiter
│   │   ├── common/                     # Contextual Missing-Data Wrapper, Net-in-
Pocket Tax Engine
│   │   ├── modules/
│   │   │   ├── indices_vix/            # NSE/BSE Indices, GIFT Nifty, Indian
ADRs, India VIX Range Calc
│   │   │   ├── equities/               # 5Y/20Q Financials, Segments, KPIs,
POC/VWAP, DCF/SIP, Peer Compare
│   │   │   ├── fno_derivatives/        # Option Chain, SciPy Greeks, Candle OI
Overlay, Mode-A 1m Replay Calc
│   │   │   ├── ipos/                   # Mainboard/SME, Restated 5Y, OFS Table,
Unlisted Price, Registrar Links
│   │   │   ├── mutual_funds/           # NAVs, Monthly Churn + AI Reasons, MF-vs-
ETF Calc, Peer Compare
│   │   │   ├── etfs/                   # Live Price vs iNAV, Impact Cost,
Tracking Error, Liquidity
│   │   │   ├── bonds/                  # G-Secs, SDLs, Corporate NCDs, Yield
Curve, Clean/Dirty & YTM Calc
│   │   │   ├── commodities/            # Gold/Silver MCX, SGB Discount Scanner,
SGB vs ETF vs Physical Calc
│   │   │   ├── reits_invits/           # Occupancy, WALE, Cap Rate, 4-Component
Distribution Tax Calc
│   │   │   ├── events/                 # Dividends, Splits, Bonuses, Buybacks,
Mergers, Rights, Earnings
│   │   │   ├── market_pulse/           # 4 PM EOD Sector Money Flow, Delivery vs
Intraday, Volume Shockers
│   │   │   ├── portfolio_sandbox/      # CAS PDF Portfolio X-Ray + ₹10L Zero-KYC
Paper-Trading Sandbox
│   │   │   ├── ai_quant_hub/           # RangeCast AI, StreetPulse Matrix, 4PM X-
Ray, Ask Dalal AI
│   │   │   └── academy/                # Fundamentals + 7 Asset/F&O Tabs + Hover-
to-Decode Tooltip API
│   │   ├── ai_engine/
│   │   │   ├── llm_router.py           # Zero-Cost Failover Router (Gemini Free -
> Groq -> OpenRouter)
│   │   │   ├── rag_pipeline.py         # Docling PDF Table Parser + pgvector
Hybrid Search
│   │   │   ├── finbert_sentiment.py    # Local CPU Sentiment Engine
(Reddit/FinTwit/News/Filings)
│   │   │   └── range_forecast.py       # LightGBM + F&O OI Walls + Z-Score 7D/14D
Band Engine
│   │   └── ingestion/                  # XBRL Parser, Broker WebSocket
Multiplexer, Celery Workers, DLQ
│   └── requirements.txt
├── frontend/                           # Next.js (App Router, TypeScript
```

7. Consolidated Master PostgreSQL Schema (Relational + TimescaleDB + pgvector)

All data resides inside a single PostgreSQL 16 instance with `timescaledb` and `vector` extensions enabled. Every financial metric column in JSONB adheres to the contextual missing-data contract: `{"val": 1420.5, "status": "REPORTED", "reason": null}`.

```
-- 1. Enable Core Extensions Inside Single PostgreSQL Instance
CREATE EXTENSION IF NOT EXISTS timescaledb;
CREATE EXTENSION IF NOT EXISTS vector;

-- 2. Master Companies & F&O Eligibility Table
CREATE TABLE companies (
    id SERIAL PRIMARY KEY,
    nseSymbol VARCHAR(30) UNIQUE,
    bseCode VARCHAR(20) UNIQUE,
    isin VARCHAR(20) UNIQUE NOT NULL,
    companyName VARCHAR(255) NOT NULL,
    sector VARCHAR(100) NOT NULL,
    industry VARCHAR(150) NOT NULL,
    listingDate DATE,
    isFnoEligible BOOLEAN DEFAULT FALSE,
    fnoLotSize INT,
    sebiFnoCriteriaJson JSONB,              -- MWPL, Quarter Sigma & Delivery
progress for Cash-only stocks
    isActive BOOLEAN DEFAULT TRUE
);

-- 3. 5-Year Annual & 20-Quarter Financials (Contextual Missing-Data JSONB)
CREATE TABLE companyFinancialsPeriod (
    companyId INT REFERENCES companies(id),
    periodType VARCHAR(10) NOT NULL,        -- 'ANNUAL' | 'QUARTERLY' | 'TTM'
    periodEndDate DATE NOT NULL,
    isConsolidated BOOLEAN DEFAULT TRUE,
    incomeStatementJson JSONB NOT NULL,     -- Revenue, ARR, EBITDA, OPM%, PAT,
EPS with status/reason
    balanceSheetJson JSONB NOT NULL,        -- Equity, Reserves, Gross/Net Debt,
CWIP, Working Capital Days
    cashFlowJson JSONB NOT NULL,            -- CFO, CFI, CFF, FCFF, FCFE,
Maintenance vs Growth Capex
    ratiosAndDupontJson JSONB NOT NULL,     -- ROE, ROCE, ROA, 5-Step DuPont,
CFO/EBITDA, PEG, Graham
    segmentsIndAs108Json JSONB,             -- Multi-domain vertical &
Domestic/International split
    operationalKpisJson JSONB,              -- Sector-specific NIM/CASA,
Attrition/TCV, ARPOB, OrderBook
    forensicGovernanceJson JSONB,           -- Piotroski F, Altman Z, Beneish M,
RPT, Contingent Liab, Ratings
    provenanceMetaJson JSONB,               -- Official XBRL/PDF filing link &
timestamp
    PRIMARY KEY (companyId, periodType, periodEndDate, isConsolidated)
);

-- 4. Indices, GIFT Nifty, Indian ADRs & India VIX Master Tables
CREATE TABLE indicesMaster (
    id SERIAL PRIMARY KEY,
    indexSymbol VARCHAR(50) UNIQUE NOT NULL, -- e.g., 'NIFTY 50', 'SENSEX',
'BANKNIFTY', 'INDIA VIX'
    exchange VARCHAR(15) NOT NULL,           -- 'NSE' | 'BSE' | 'NSE_IX' |
'US_ADR'
    category VARCHAR(30) NOT NULL,           -- 'BENCHMARK' | 'SECTORAL' |
'THEMATIC' | 'VOLATILITY' | 'PREMARKET'
```

8. Iterative 6-Sprint SDLC Execution Plan

Sprint	Core Modules Delivered	Engineering Milestones & Verification Criteria
Sprint 1 Foundation & Academy	Docker Stack, PostgreSQL (`TimescaleDB` + `pgvector`), Redis, `Market Academy` (`/learn`), Hover Tooltips, Net-in-Pocket Tax Engine	Initialize Modular Monolith FastAPI + Next.js shell; seed `Fundamentals` and all 7 asset/F&O learning tabs; activate universal Hover-to-Decode tooltips and statutory brokerage/tax calculator utility.
Sprint 2 Equities & Peer Engine	Equities 5Y/20Q Deep-Dive (`/stocks/[symbol]`), XBRL Parser, Volume Profile/VWAP Charts, DCF/SIP Calculators, `+ Compare with Peers`	Build contextual missing-data wrapper, Standalone/Consolidated & Common-Size toggles, Ind AS 108 segments, Piotroski/Altman/Beneish scores, and lazy-loaded Peer Comparison drawer with persistent session state.
Sprint 3 Indices, VIX & F&O Hub	Indices Hub (`/indices`), GIFT Nifty & ADR Gap Engine, India VIX Stability, Stock/Index Option Chain (`/fno`), Mode-A 1m Option Replay	Implement NSE Tuesday / BSE Thursday expiry calendar, SciPy Black-Scholes Greeks (IV, Delta, Gamma, Theta, Vega, POP), Candlestick Call/Put OI overlays, and Mode-A 1-Minute Historical "What-If" Replay & Theta-Attribution Calculator.
Sprint 4 IPOs, MFs, ETFs & Events	IPO Hub (`/ipos`), Mutual Funds (`/mutual-funds`), ETFs (`/etfs`), Corporate Events Calendar (`/events`)	Deliver DRHP 5Y restated financials, OFS exit table, Unlisted Price comparison, T+1 Registrar links, Monthly MF Churn + Two-Tier AI reasons, MF-vs-ETF True Exit Cost calculator, and all 5 Corporate Event calculators.
Sprint 5 Bonds, Gold, REITs & Sandbox	Bonds (`/bonds`), Commodities (`/commodities`), REITs (`/reits-invits`), 4 PM Market Pulse, Portfolio X-Ray & ₹10L Paper-Trading Sandbox	Launch Sovereign Yield Curve, Clean/Dirty Bond calculator, Live 60+ SGB Discount Scanner, REIT 4-part tax calculator, 4 PM Delivery vs. Intraday flow map, and Zero-KYC ₹10 Lakh Virtual Paper Trading.
Sprint 6 AI Quant Hub & Cloudflare	`AI Quant Hub` (`RangeCast AI`, `StreetPulse Matrix`, `4 PM X-Ray`, `Ask Dalal AI`), Zero-Cost LLM Failover Router, Cloudflare Tunnel	Connect Gemini/Groq/OpenRouter failover router, `Docling` DRHP/Concall RAG, CEO Say-vs-Do tracker, and GitHub CI/CD to Cloudflare Pages & Zero-Trust Tunnel on custom domain.

9. Instructions for AI IDE & Autonomous Coding Agents

When this specification document is provided to an AI IDE (such as Cursor, Windsurf, GitHub Copilot Workspace, or Claude Code) or a software engineer to execute the build:

1. **Treat Functional & Mathematical Specifications as Mandatory:** Every module, metric, contextual missing-data reason, statutory charge deduction, SEBI Tuesday/Thursday expiry rule, and calculator (including Mode A 1-Minute Option Replay and MF-vs-ETF Exit Cost math) defined in Sections 1 through 8 must be implemented accurately without placeholder stubs.
2. **Exercise Full Architectural Autonomy on Code Organization:** You have complete permission to adapt the folder tree, split large service files into smaller sub-modules, add intermediate database join tables or indexes, or adjust library versions so the codebase remains clean, modular, and bug-free during incremental sprint development.